NOT MEASUREMENT SENSITIVE

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MILITARY SPECIFICATION

# TRUCKS, PIPE AND SEWER CLEANER: HIGH PRESSURE WATER JET TYPE, 5680 LITER (1,500 GALLON), 12 700 KILOGRAMS (28,000 POUNDS) GVW, 4x2, MODIFIED COMMERCIAL

This specification is approved for use by the US Army Tank-Automotive Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers diesel and gasoline engine driven, four-wheel, two-rear-wheel drive, 5680 liter (L) (1,500 gallon), high pressure water jet, pipe and sewer cleaner modified commercial trucks having a gross vehicle weight (GVW) of 12700 kilograms (kg) (28,000 pounds), for normal operating conditions. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| | W-B-131 | - Battery, Storage: Vehicular Lighting and Starting. |
| * | VV-F-800 | - Fuel Oil, Diesel. |

MILITARY

| | | |
|---|---|---|
| * | MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |
| * | MIL-T-5624 | - Turbine Fuel, Aviation, JP-4 and JP-5. |
| * | MIL-M-43719 | - Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, General Specifications For. |
| * | MIL-T-83113 | - Turbine Fuel, Aviation, Kerosene Type, Grade JP-8. |
| | MS 51118 | - Pintle Assembly, Towing- 40,000 Lbs. Capacity, Manual Release. |
| | MS 51317 | - Light, Warning, Vehicular, Rotating, D.C. |

STANDARDS

FEDERAL

| | | |
|---|---|---|
| | FED-STD-297 | - Rustproofing of Commercial (Nontactical) Vehicles. |

MILITARY

| | | |
|---|---|---|
| | MIL-STD-1223 | - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards. |

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.



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* DEPARTMENT OF DEFENSE (DOD)
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DOT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

* (Application for copies of DOT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington. D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM D 4985 - Standard Specification for Low Silicate Ethylene Glycol Base Engine Coolant for Heavy Duty Engines Requiring an Initial Charge of Supplemental Coolant Additive.

(Application for copies of ASTM publications should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual



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(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

SAE, INC.

SAE Standards and Recommended Practices

| | | |
|---|---|---|
| | J318 | - Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers -Trucks. Truck-Tractors, and Trailers. |
| * | J350 | - Spark Arrester Test Procedure for Medium Size Engines (DOD Adopted). |
| | J537 | - Storage Batteries. |
| | J551 | - Limits and Methods of Measurement of Radio Interference Characteristics of Vehicles and Devices (30-1000 MHz). |
| | J560 | - Seven Conductor Electrical Connector for Truck Trailer Jumper Cable. |
| | J588 | - Turn Signal Lamps. |
| | J589 | - Turn Signal Switch. |
| | J682 | - Rear Wheel Splash and Stone Throw Protection. |
| | J683 | - Tire Chain Clearance. Trucks. Buses (Except Suburban, Intercity, and Transit Buses), and Combinations of Vehicles. |
| * | J688 | - Truck Ability Prediction Procedure. |
| | J704 | - Transmission Mounted Power Take-Offs. |
| | J844 | - Nonmetallic Air Brake System Tubing. |
| | J1067 | - Seven Conductor Jacketed Cable for Truck and Trailer Connectors. |
| * | J1349 | - Engine Power Test Code. Spark Ignition and Diesel. |

(Application for copies of SAE publications should be addressed to SAE, INC., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.

Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the document. These documents also may be available in or through libraries or other informational services.)



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* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.12, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.12, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturers technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Markings and data plates. As specified by the procuring activity for the appropriate service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with MIL-STD-1223. When specified (see 6.2), tropical rustproofing in accordance with MIL-STD-1223 shall be furnished.

* 3.1.1.4 Drain plugs. Drain plugs installed in manual transmissions and rear axles shall be of the permanent magnetic type.

* 3.1.1.5 Towing devices. Towing devices consisting of two hooks, loops, eyes or pins or the chassis manufacturer's standard single center mounted eye or pin shall be mounted on the front of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.



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* 3.1.1.6 Wheel splash and stone throw protection. The vehicle shall have rubber mud flaps to the rear of the rear wheels. All tilt cabs shall have rubber mud flaps to the rear of the front wheel. A metal strip not less than 3.2 millimeters (mm) (0.125 inch) thick and not less than 25 mm (1 inch) wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.25) inch wide by 2.4 mm (0.094 inch) thick shall be furnished at each bolt. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE 3682.

* 3.1.1.7 Trailer towing package. When specified (see 6.2), a trailer towing package consisting of a pintle, safety chain attachment devices, a lighting receptacle and associated reinforcements and wiring shall be installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS 51118 and shall be installed on the chassis frame with reinforcements to transfer a vertical tongue load of not less than 815 kg (1,800 pounds) and a horizontal drawbar load of not less than 80 kilonewtons (kN) (18,000 pounds) directly to the chassis rails. The rearmost portion of the pintle shall be forward but not more than 100 mm (4 inches) forward of the rearmost part of the vehicle. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the truck furnished. The lighting receptacle, conforming to SAE J560 with its conductors connected and color coded as specified therein, shall be mounted in a readily accessible location near the pintle.

3.1.1.8 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with an exhaust system spark arrester.

* 3.1.1.9 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

* 3.1.1.10 Hydraulic fluid identification plate. When a body hydraulic system is furnished and the hydraulic system requires fluid replenishment, a decal or plate shall be installed near the filler cap and shall identify the type of hydraulic fluid to be used. (See 3.1.1.11.)

* 3.1.1.11 Decals and data plates. The hydraulic fluid identification information and the power takeoff caution notice shall be on a standard decal or plate from the supplier of that item or shall conform to type I or type II, class 1 of MIL-M-43719 or composition A (class l or 2) or composition C of MIL-P-514.



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* 3.1.1.12 Tool box. When specified (see 6.2), a tool box shall be furnished. The tool box shall provide for storage in addition to that required by 3.4.16. Minimum dimensions shall be 560 mm (22 inches) by 560 mm (22 inches) by 300 mm (12 inches). A door opening size of not less than 510 mm (20 inches) by 255 mm (10 inches) shall be furnished. The tool box shall be fabricated of not less than 2.657 mm (12 gage) (0.1046 inch) steel or equivalent strength aluminum. The box shall be weatherproof and shall provide for locking with a padlock. The tool box shall be mounted to the immediate rear of the cab, on the curbside of the vehicle.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with EPA regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The cab interior sound level shall not exceed 84 db(A) when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; the body; and a full complement of fuel, lubricants, and coolant.

3.2.5 Gross vehicle weight. The gross vehicle weight (GVW) shall consist of the curb weight, the operator weight (computed at 80 kg (175 pounds)), and a payload to provide not less than the specified GVW.

3.2.6 Weight distribution. The distribution of GVW for the purpose of establishing suspension, axle and tire capacities shall be determined with the payload uniformly distributed over the load area.

* 3.2.7 Ratings. Vehicle ratings shall be manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the engineering office of the procuring activity. Minimum GVW rating shall be 12700 kg (28,000 pounds).



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3.2.8 Cab-to-axle. The vehicle cab-to-axle dimension shall no be less than 2130 mm (84 inches).

* 3.2.9 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.10 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with truck loaded to specified GVW.

* 3.3.1.1 High speed gradeability. The vehicle shall ascend continuous grades of 2.0 percent at 80 kilometers per hour (km/h) (50 miles per hour (mph)). Gradeability requirements shall be met with the main transmission in direct drive and, when multispeed axle is furnished with the axle in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

* 3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be calculated with engine operating at not less than 35 percent of recommended governed speed, and shall provide a vehicle speed of not more than 7.2 km/h (4.5 mph).

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 93 km/h (58 mph). Conformance to geared speed specified shall be determined by calculations made in accordance with the following formula:

$$\text{Maximum geared speed (km/h)} = \frac{\text{Governed speed (rpm) X 1.609}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.



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* 3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine. Engine net power used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristic specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

* 3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed of not more than manufacturer's recommended operating speed. The engine shall be capable of continuous, warranted operation on unleaded fuel, when used in accordance with the operator's manual. Net power used in performance prediction calculations shall be determined in accordance with SAE J1349.

* 3.4.1.3 Oil filter. A full flow or combination full flow and bypass type oil filter with replaceable element shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to GVW and operated at an altitude of 3050 meters (m) (10,000 feet) above sea level or in an ambient air temperature of not less than 52° Celsius (°C) (125 degrees Fahrenheit (°F)). A coolant level indicator consisting of an unbreakable translucent reservoir shall be provided in a position readily visible for checking the coolant level. The translucent reservoir shall be part of the coolant recovery system. The cooling system shall be of sufficient capacity to permit idling the engine (including high idle at full air conditioning load) in an ambient temperature of 35°C (95°F), for a period of not less than one half hour without overheating. For cooling system servicing, see 3.4.24.1.

* 3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.



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* 3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

3.4.1.8 Silicone rubber hoses. When specified (see 6.2), silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.9 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and fuel warmers (diesel engine driven vehicles only) shall be provided. Heaters shall operate on 110 volts alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding the vehicle. The receptacle shall be located on the streetside of the vehicle, as near the cab door as practicable. A three-wire connecting cable, not less than 7600 mm (25 feet) long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle end and a standard, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). The electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to -51°C (-60°F)) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters and fuel warmers shall be furnished as follows:

(a) A coolant heater, 1,000-watt (W) minimum rating on trucks up to class 22 and 1,500-watt minimum rating for larger trucks shall be installed in the engine block or in the lower coolant inlet hose. An engine thermostat with an operating range of 77°C (170°F) to 90°C (195°F) shall be installed.

* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:
  (1) Not less than 16 W/L (15 watts per quart) heating capacity
  (2) Not more than 2.8 watts per square centimeter ($W/cm^2$) (18 watts per square inch) heating capacity
  (3) Thermal balance design or thermostat control providing for uninterrupted operation
  (4) Provision for mounting below minimum service oil level.

* (c) An in-line fuel warmer or preheater unit shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The unit shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°C (-40°F) to an outlet temperature of -13°C (+9°F), with a fuel flow rate not less than the maximum fuel demand of the engine fuel



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system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit. The unit shall not cause heating of the fuel above 27°C (80°F) under any possible condition.

* (d) An in-tank fuel warmer or preheater unit shall be provided on diesel engine driven vehicles. The unit shall use engine coolant to transfer heat to the fuel in one fuel tank. The warmer shall not cause heating of any fuel above 27°C (80°F) under any possible condition, shall not disable or cause elimination of the fuel gage sending unit and shall not violate 3.2.1 or any Federal Motor Carrier Safety Regulation. A coolant shut off valve shall be included. The units required by 3.4.1.9(c) and (d) may be combined.

* 3.4.1.10 Fuel fired engine preheater. When specified (see 6.2), for diesel engine driven vehicles, a diesel fuel fired engine water heater shall be furnished to preheat the engine. The heater shall include a timer, a thermostat and a circulating pump, and shall be connected to the engine coolant system. The heater shall be capable of starting and operating at -51°C (-60°F) and shall heat the engine to +4°C (+40°F) from -51°C (-60°F) in not more than 1 hour. The system shall be equipped with a light, visible to the driver, to indicate that the preheater is operating. The system shall include fuel warmers or preheaters conforming to 3.4.1.9(c) and (d).

* 3.4.1.11 Vernier throttle control. When specified (see 6.2), a manual vernier controlled throttle with quick release shall be furnished, except on electronically controlled engines.

3.4.2 Electrical systems. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. A 12- or 24-volt direct current (dc) starting system with 12-volt dc lighting system shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or electric grid heater. If an ether system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 355 milliliters (mL) (12 fluid ounces) shall be furnished. A reservoir need not be furnished. When specified (see 6.2) the starting motor shall be equipped with a thermostat controlled, automatic resetting circuit breaker that will protect the motor from overcrank heat damage. The circuit breaker shall not limit cranking ability for a startable engine.

3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished.



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* 3.4.2.3 Alternator. Unless otherwise specified (see 6.2) on diesel engine driven vehicles, a minimum 15-ampere alternator shall be furnished. The alternator output with the engine at engine idle speed shall be not less than 45 amperes. Unless otherwise specified (see 6.2) on gasoline engine driven vehicles, an alternator of not less than 85 amperes rated capacity, and which provides not less than 45 amperes dc output at normal engine idle speed, shall be furnished.

3.4.2.4 Lighting. All vehicle lights, reflectors, and wiring shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails (unless recessed and fully protected) or mounted on vehicle bumpers.

* 3.4.2.5 Turn signal. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. The turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall have a visible flash indicator.

* 3.4.2.6 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity ratings and the total cold cranking ampere ratings at -18°C (0°F), both measured in accordance with SAE J537, shall be not less than specified in table I. The batteries shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131.

* TABLE I. Batteries.

| Engine type | Reserve capacity (minutes) | Cold cranking (amperes) |
|---|---|---|
| Diesel | 480 | 1,875 |
| Gasoline | 100 | 450 |

* 3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tank(s). The fuel tank shall be not less than 189 L (50 gallon) total capacity 163 L (43 gallons) for vehicles destined for California)). When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, a selector valve



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connecting either tank to the engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single fuel gage; or an equalizing pump shall be used to maintain the same fuel level in both tanks.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. Exhaust mufflers, when stacked, shall be furnished with a heat shield and a hinged rain cap.

* 3.4.4.1 Spark Arrester. When specified (see 6.2), a spark arrester conforming to the material and performance requirements of SAE J350 shall be furnished, except for turbocharged engines. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance

* 3.4.5 Transmission. Unless a manual transmission is specified, the vehicle shall be equipped with an automatic transmission. When specified (see 6.2), the vehicle shall be equipped with a manual transmission, complete with clutch. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission and axle gear ratios shall be selected to provide the performance specified in 3.3.1 through 3.3.1.3.

* 3.4.5.1 Automatic transmission. The automatic transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with a power takeoff opening. The net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine.

* 3.4.5.2 Manual transmission. The manual transmission and other vehicle equipment shall provide not less than five forward vehicle speeds. Gear ratios in the transmission and axle shall be matched to provide a progressive shifting pattern throughout the complete range. The transmission shall be provided with SAE J704 power takeoff openings at the right and left sides. The transmission shall provide for maximum ease of shifting in all speeds.

* 3.4.5.2.1 Clutch. The clutch shall be the largest capacity clutch offered for the type and class vehicle and engine furnished, with the clutch torque capacity exceeding the maximum delivered engine torque. The clutch lining shall be asbestos free.

3.4.5.2 Power takeoff. When a power takeoff is furnished, it shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading “DO NOT OPERATE



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VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be installed so as to be readily visible to the seated driver.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The vehicle shall have a heavy duty main frame or frame reinforcements. The heavy duty frame shall have frame rails of greater section modulus than the manufacturer's standard for the vehicle furnished and shall provide structural strength at least equivalent to the reinforced frame specified herein. Reinforcements shall extend at least from the rear of the front-suspension, rear hanger bracket to the front of the rear-spring, front hanger bracket.

* 3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to specified GVW. When the suspension is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels. When specified (see 6.2), auxiliary rear springs shall be provided.

* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the vehicle loaded to specified GVW. The wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Two speed axle. When specified (see 6.2), a two-speed axle shall be furnished, equipped with electric, vacuum, or air shift and provided with ratios which will permit proper gear splitting. Gear ratio shall provide performance specified in 3.3.1 thru 3.3.2.

* 3.4.9.2 Traction control. When specified (see 6.2), a traction control shall be furnished. The traction control shall actuate automatically to ensure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Except when tube type tires are specified in acquisition documents, multi-piece rims shall not be furnished. Tire, rim, and disc wheel sizes shall be the same for all wheels on each vehicle. Tire



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size and load range (ply rating) shall be the same for all tires on each vehicle. When specified (see 6.2), 10 stud disc type wheels shall be furnished.

* 3.4.10.1 Tires. Steel belted radial or, when specified (see 6.2), bias ply tires shall be furnished. Unless otherwise specified, tires shall be tubeless type with highway tread. Unless otherwise specified, low profile or standard profile tires may be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with vehicle loaded to specified GVW. Tires shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.4.10.2 Inner tubes. When tube type tires are specified in procurement, inner tubes shall be of heavy duty type, and shall be of proper size for tires furnished. Tube type radial tires shall have radial tire inner tubes. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable safe removal or mounting of a spare wheel assembly using only the tools specified in 3.4.16.1. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished.

* 3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the front axle of the vehicle.

* 3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52. Brake linings shall be of nonasbestos material.

3.4.11.1 Service brakes. The vehicle shall be equipped with full-air brakes on all wheels. The braking system, complete with necessary components, shall include:



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(a) Air compressor, unloader-head-type, engine driven and engine lubricated, air or water cooled, and having a capacity of not leas than 205 L/min (7.250 cubic feet per minute (cmf));
(b) Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank;
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low-air-pressure warning, visible and audible
(g) Service brake stop lamp switch
* (h) Automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes, on all axles.
(h) Automatic moisture ejector.

* 3.4.11.1.1 Air driver. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

* 3.4.11.2 Trailer brake control system. When specified (see 6.2), in addition to the components specified in 3.4.11.1, a trailer brake control system shall be furnished. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control
(c) Independent hand control for trailer brakes
(d) Prime mover protection valve with dash control and automatic breakaway feature
(e) Trailer stoplight control operative with foot brake and with hand control for trailer brakes
(f) Two SAE J844 coiled air hoses, not less than 2800 mm (110 inches) long when fully extended, with SAE J318 gladhand couplers an both ends of hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment;
(g) Air connectors for trailers with SAE J318 gladhand couplers mounted at the rear of the vehicle located to prevent interference with trailer
(h) Dummy gladhand couplers with chains
(i) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system.

3.4.11.2.1 Brake control from a towing vehicle. When specified (see 6.2), the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brake is furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position



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providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional full width cab. When specified (see 6.2), a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism, shall be furnished. Tilting shall not interfere with present or future installations of additional equipment, such as a front mounted winch. When specified (see 6.2), a tilt type cab with tilting and locking mechanism shall be furnished. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 610 mm (24 inches). Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with both doors equipped with external, key operated locks. Drip rails shall be installed above the cab doors. Safety grips or grab handles shall be provided on each side of the cab to assist personnel in entering and leaving the cab. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be furnished. For tilt type cabs, provisions to facilitate cleaning the windshield shall be provided by means of a bumper step, or bumper step cutouts, and a grab handle located under the windshield.

* 3.4.12.1 Cab interior. Unless otherwise specified, the cab shall have an upholstered, full width, adjustable seat and back or individual, adjustable, driver's seat and individual passenger seat. When specified (see 6.2), an individual, adjustable, driver's seat and an individual passenger seat shall be provided. When specified (see 6.2), the driver's seat shall be the manufacturer's standard air ride suspension type. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Seat belts shall be installed for each seating position. Outboard seats shall have combination pelvic and upper torso restraint seat belts. Cab doors shall have windows with crank-operated window regulators. A rear window shall be provided. Interior lighting shall be provided.

3.4.13 Steering. Power steering shall be furnished.

* 3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by either air or electric motor(s). When specified (see 6.2), and if electric motor wipers are furnished, intermittent wipers shall be furnished.



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3.4.15 Bumpers. Unless the bumper is an integral part of the vehicle cab, a channel type front bumper shall be provided.

3.4.15.1 Rear end protection. The rear of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86. A rear bumper shall be provided.

* 3.4.16 Tool storage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains (for outside tires on duals only), and emergency reflective triangles shall be furnished. The stowage space shall provide for positive retainment of this equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of the cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.16.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under an axle, or other satisfactory lift point at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit the removal and replacement of the wheel and tire assembly.

* 3.4.17 Heater and defroster. The vehicle shall be provided with a hot water heater with fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to windshield defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.18 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. All instruments shall be visible to the driver when seated in the driving position.

* 3.4.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key-operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
* (d) Oil pressure gage or red indicator warning light



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* (e) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
* (f) High coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Ash receptacle
(i) Dual sunvisors
(j) Driver's compartment ventilator other than window
(k) Tachometer.

When specified (see 6.2), an ammeter or voltmeter gage, oil pressure gage, and coolant temperature gage shall be provided in lieu of the indicators and buzzers specified above.

* 3.4.20 Rearview mirrors. Outside rearview mirrors shall be mounted on both sides of the cab. The mirrors shall have flat and convex areas. The flat portion shall have not less than 320 square centimeters ($cm^2$) (50 square inches) of reflective area. The convex portion shall have not less than 155 $cm^2$ (24 square inches) of reflective area and a radius of curvature of not less than 510 mm (20 inches). The convex portion shall be attached to the lower mirror supporting arm and shall not interfere with use of the flat mirror. The mirrors shall have not less than two supporting arms. When specified (see 6.2), the curbside flat mirror shall be of the motorized type, with remote control. The mirror motor shall provide not less than 60 degrees horizontal rotational viewing range. When specified (see 6.2), the flat mirrors shall be electrically heated. Mirror remote and heating controls shall be within reach of the seated driver.

* 3.4.21 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, and air operated horn shall be furnished.

3.4.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.23 AM/FM radio. When specified (see 6.2), the manufacturer's standard AM/FM radio shall be provided.

3.4.24 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights, adjustment of the engine and brake systems; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine and running gear withgrades o£ lubricants recommended



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for the ambient air temperature at the delivery point; servicing of the cooling system in accordance with 3.4.24.1; and servicing of the windshield washer reservoir with water and appropriate additives.

* 3.4.24.1 Engine coolant. The engine coolant shall be a solution of ethylene glycol antifreeze and water or propylene glycol antifreeze and water, in equal parts of antifreeze and water by volume. Ethylene glycol antifreeze shall conform to ASTM D 4985 with not more than 250 parts per million silicates. Propylene glycol antifreeze shall be in the specific formulation approved by the engine and cooling system manufacturers. When specified (see 6.2), the percentage of antifreeze in the cooling system shall be increased to provide protection against freezing down to -54°C (-65°F).

3.5 Body. The body of the vehicle shall consist of a completely equipped pipe and sewer cleaning unit of the high pressure, water-jet type. Components comprising the body assembly shall include a water supply tank, a water pump (power takeoff or auxiliary engine driven), a high pressure water system, a hose reel, and various associated equipment as specified herein. The body shall be totally inclosed with adequate number of doors provided to permit access to the engine, pumps, and operating components. A rear canopy, or door that when fully open locks into a canopy position, shall be provided.

* 3.5.1 Body skirting. The body shall be of full skirted design. The skirting shall extend down to approximately the center of the wheels and shall be provided on each side. The skirting shall incorporate three or four underslung storage compartments, one on each side in the fore part of the body skirting, and one or two in the aft part. Total cabinet inclosed volume shall be not less than 999,600 cubic centimeters ($cm^3$) (61,000 cubic inches). Compartment doors shall be equipped with non-corrosive hinges and recessed paddle type locks. All cabinet locks shall be keyed alike.

3.5.2 Tank. The tank shall be of not less than 5680 L (1500 gallon) capacity, and shall have not less than six sections. The bottom 20 percent of the tank shall remain unbaffled. The length of the tank shall provide proper load distribution on the front and rear axles of the chassis. The tank shall have an elliptical or modified rectangular cross section. The tank shall be fabricated from low-alloy steel. The tank shell shall be not less than 10 gage (3.416 mm) (0.1345 inch) thickness. The tank heads shall be not less than 10 gage (3.416 mm) (0.1345 inch) thickness and shall be dished or otherwise constructed to obtain equivalent strength. The tank shall be of watertight, welded, smooth skin construction and shall withstand 20.7 kPa (three pounds per square inch (psi)) pressure without leakage. The interior surfaces of the tank (including the top) and the external surface (at the bottom) shall be coated with a fire cured baked epoxy. Access opening(s) at the top of the tank, for purposes of inspection and cleaning, shall be furnished.



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3.5.2.1 Tank connections. The pipe connection to the pump shall be located in such a position that the entire bottom of the tank serves as a low velocity settling chamber. A 50 mm (two-inch) drain fitting equipped with a 50 mm (two-inch) gate valve, located for easy access, shall be provided in the bottom or at the rear so water can be dumped directly into a manhole without moving the machine. External sight gages or means of determining when the tank is full shall be provided on each side. The external sight gages or other means shall be located so that they are visible both during the filling operation and the cleaning operation. The tank filler mechanism shall allow the tank to be filled from both the curb side and the street side of the vehicle. Tank filler mechanism shall include a 100 mm (four-inch) air gap. The tank shall be vented at all times to prevent the buildup of either positive or negative pressure within the tank from any cause. Access opening(s) at the top of the tank, for inspection and cleaning purposes shall be furnished. A permanently attached steel ladder to provide access to the top of the tank shall be provided.

3.5.3 Auxiliary engine. Unless a power takeoff is furnished, an auxiliary engine, operating on the same type of fuel as the main chassis engine, shall be provided for driving the water pump. The auxiliary engine shall be a liquid cooled, four-stroke cycle, (two or four-stroke cycle for diesel engines) reciprocating industrial type. The engine furnished shall be capable of meeting the performance requirements when operated at an engine speed no greater than the manufacturer's recommended operating speed. When a gasoline engine is furnished, it shall have not less than a 4920 $cm^3$ (300 cubic inch) displacement.

3.5.3.1 Auxiliary engine mounting and enclosure. The auxiliary engine shall be totally enclosed and fitted with ventilation panels and hood that shall prevent road splash and rain. Hinged access panels, constructed with louvers for ventilation, shall be installed on each side and secured with a positive locking device. Hinges shall be bolted to cabinet doors. The panels and doors shall not vibrate or drum. The panels and doors shall be constructed of not less than 16 gage (1.519 mm) (0.0598 inch) hot rolled mild steel.

* 3.5.3.2 Engine control devices. Control devices shall be furnished for the engine (see 3.5.6). Automatic control devices shall include a high temperature control (coolant control set at 104°C (220°F)), an oil pressure control set at the engine manufacturer's minimum recommended oil pressure.

3.5.3.3 Starting system. A starting system of 12-volt dc potential shall be used on the auxiliary engine. The system shall include a starting motor with solenoid switch, an alternator, and a voltage regulator. A battery of 12-volt potential, with capacity compatible with the application, shall be furnished.



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3.5.3.4 Engine components. The engine fuel supply shall be provided by the truck chassis fuel tank. A fuel pump and filter shall be provided. The muffler shall be positioned so that the fumes do not interfere with the operator.

3.5.4 Water pump. The water pump shall be of the positive displacement reciprocating type. The pump shall be capable of delivering a minimum of 227 L/m (60 gallons per minute (gpm)) against not less than a 540 790 kPa (1650 psi) head at 500 revolutions per minute (rpm) maximum. The drive shall be multiple "V" belt or direct from the power takeoff, and shall be of adequate size to transmit the power from the auxiliary engine to the pump without slippage. When a multiple "V" belt is furnished, a screw jack type tensioning device shall be provided on the engine so that proper tension can be maintained in the belts. Positive means to prevent total shutoff of the outlet of the pump shall be provided. A multi-purpose valve shall be incorporated. This valve, in the proper position, shall direct the flow of water to the cleaning nozzle under increased pressure or provide water to the wash down hose or direct the water to the recirculatory system as required.

3.5.4.1 Piping. All high pressure piping, both pipe and fittings, shall be of steel and shall be adequate for continuous operation at pressures up to the maximum output pressure of the water pump with adequate provision for over pressure. Piping shall have a safety factor of not less than 4 to 1. All suction and low pressure return lines may be of standard weight pipe and malleable fittings. Cast iron fittings shall not be used. The piping system shall be designed so as to facilitate complete draining of the entire system.

* 3.5.5 Hydraulic drive system. The hydraulic drive system that powers the hose reel shall consist of a five gallon tank, a return line filter with replaceable element, a direction and speed control valve with built-in overload relief, a dual counterbalance valve to provide hydraulic shock relief, a speed control valve, high pressure hoses with integral swivel type fittings, a pump driven by the power takeoff or the auxiliary engine and a reversible hydraulic motor. The hydraulic motor shall be of the gearotor type capable of delivering at least 960 Newton meters (N m) (8500 inch pounds) of torque at the hose reel shaft.

* 3.5.6 Controls and control panel. All gages, switches, levers and control valves necessary for the control of the operation, shall be grouped together at the operator's station so that the operator shall have complete control of the cleaning operation at all times. A fixed control panel shall be provided and shall contain as a minimum a choke (gasoline engine only), a vernier-type throttle, an ammeter, an engine water temperature gage, an engine oil pressure gage, a low water warning light, a key type combination ignition starter switch, a reel speed control, a reel hydraulic system pressure gage, a water system pressure gage and a quick disconnect type of connection for the wash down hose. Convenient access to the rear side of the controls and gages mounted on the



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fixed control panel shall be furnished. The operator shall be completely shielded from all the apparatus and high-pressure piping. The noise level in the operator's area shall not exceed 84 db(A).

3.5.6.1 Portable control. In addition to the controls and gages attached to the fixed control panel, a portable control station, that shall be capable of being detached from the fixed station and carried by the operator, shall be furnished. The portable control station shall be linked to the fixed control panel by a multi-conductor flexible electrical cable and shall contain two switches: one switch to turn the water on or off; the other to raise, lower or stop the nozzle and cleaning hose. The portable control station shall enable control of operation of apparatus from any location within at least 15 m (50 feet) from the fixed station.

3.5.7 Hose reel. The hose reel shall be of heavy duty construction and shall be capable of withstanding any mode of operation without collapsing or distorting. The ends shall be rounded to prevent damage to the hose when it comes into contact. The reel shall be capable of holding 305 m (1,000 feet) of 25 mm (one inch) inside diameter (id) high pressure plastic hose. The reel shall be power driven in both directions. The reel shall be equipped with a manual means of emergency operation. The reel shall be designed so that it will not be necessary for the operator to guide the hose on or off. All drive chains, sprockets, and like items shall be provided with guards where necessary for the safety of the operator.

3.5.7.1 Hose. Not less than 185 m (six hundred feet) of 25 mm (one inch) id heavy duty plastic hose, shall be furnished and installed on the reel. The hose working pressure shall be not less than the maximum output pressure of the water pump. The hose shall be one-piece. The cover of the hose shall be resistant to the scuffing, acids and greases normally encountered during the cleaning of a sewer line. The hose and fittings shall be capable of withstanding a minimum of 13340 Newtons (N) (3,000 pounds) tensile load without damage. The length of the hose shall not change more than one percent as a result of changes in internal pressure in the range of 0 kPa (0 psi) to 17 790 kPa (2,000 psi) at the inlet end of the hose. The end of the hose shall be provided with a fitting to accept lead hoses, nozzle and cleaning gun assemblies specified in 3.5.7.2 and 3.5.7.4.

3.5.7.2 Nozzles. Stainless steel or hardened carbon steel cleaning nozzles shall be furnished. A 15 to 18 degree radial nozzle and a 30 to 35 degree radial nozzle shall be provided.

3.5.7.3 Hose guide shoe. A hose guide shoe with sectional (add-on type) handle shall be furnished. The design of the guide shoe shall facilitate entry of the nozzle and hose into sewers and shall protect the nose from the sharp ends of sewer tile.



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* 3.5.7.4 High pressure cleaning gun. A high pressure, general purpose cleaning gun shall be provided. The gun capacity shall be 13.25 to 102 L/m (3.50 to 27 gpm) within a pressure range of 1379 to 5515 kPa (200 to 800 psi). A high pressure hose 7600 mm (25 feet) in length and 13 mm (0.50 inch) id shall be provided for use with the gun. The hose shall be equipped with fittings for connection to the gun and to a high pressure water outlet.

* 3.5.7.5 Containment system. The hose and hose reel shall be provided with a containment system. The containment system shall consist of an upper guard constructed entirely of extruded transparent polycarbonate thermoplastic. The polycarbonate thermoplastic shall have a minimum impact strength of 3200 Joules per meter (J/m) (60 foot pounds per inch) and shall be ultra violet (UV) stabilized for protection against yellowing. In addition, it shall, as a minimum, have a Rockwell hardness of M70, a tensile modulus of 2,068,410 kPa (300,000 psi) and a shear modulus of 758 420 kPa (110,000 psi). The entire upper guard shall be transparent to permit viewing the hose reel and sewer hose. The containment system shall easily retract in one continuous motion around the reel, without removing any parts, and shall extend past the reel edge not more than 25 mm (1 inch). The guard shall be hinged at the top and shall have a locking mechanism for locking the guard in the work position. In addition, a hose which will fit over the sewer hose shall be furnished and shall extend from within the upper guard to the manhole lip. The entire containment system shall be constructed so as to contain the sewer hose and couplings within should a burst occur.

3.5.8 Warning light. An amber rotating, vehicular warning light conforming to MS 51317-4 shall be furnished. The light shall be mounted with reinforcement on vehicle centerline for 360 degree visibility. The warning light shall be operated by means of a separate switch with a red indicator light located on the instrument panel in the cab, and marked to indicate the function of the switch.

3.5.9 Body mounting. The body shall be secured with U-bolts, twin studs, or brackets, and shall include breaker strips. At least the most forward pair of mounting devices shall be spring loaded to permit vertical movement between the chassis and body.

3.5.9.1 U-bolts or twin studs. When U-bolts or twin studs are used, there shall be not less than three U-bolts or twin studs per side. Each U-bolt or twin stud shall have a 14 mm (0.5625 inch) diameter body with 16 mm (0.6250 inch) minimum thread diameter. Tie plates shall be at least 13 mm (0.50 inch) thick. The vehicle chassis frame shall be braced, using wood blocks at each mounting point unless the mounting point is located at a full-depth, frame crossmember. Blocks shall incorporate a keeper strap or groove for the mounting bolt and shall be of a width and thickness to assure retention. Two tie-back straps shall be provided, one on each side of the rear portion of the body sub frame to maintain body alinement on the vehicle chassis. Forward body mounting bolts shall be located to the rear of the tapered portion of the breaker strips (see 3.5.9.3).



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3.5.9.2 Brackets. When brackets are used, they shall be bolted to the web of the chassis frame rails. The body mounting brackets shall provide a means of drawing down the body on the chassis rails and provisions shall be made to prevent lateral shifting of the breaker strips. When additional holes are required to secure mounting brackets to the chassis frame rails, they shall be located within the area of the rail which is designated as being safe for drilling in accordance with the chassis manufacturer's body builders layouts. Attachments shall not interfere with nor obstruct existing chassis components.

3.5.9.3 Breaker strips. A hardwood or dense southern yellow pine breaker strip not less than 27 mm (1.06) inches finished thickness, shall be installed between longitudinal sills and the vehicle chassis frame. Breaker strips shall have a taper of 25 mm (1 inch) in 460 mm (l8 inches) at the forward end.

3.6 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

## 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the



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contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

* 4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity to the contract.

* 4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight, and distribution of curb weight on front and rear axle. The imposed loading on front and rear axle will be computed using the curb weight, and the payload, (see 3.2.5 and 3.2.6), to provide the GVW. Calculated imposed loads on front and rear axle will be utilized to ascertain that the suspension, axles and tires furnished are of adequate capacity to meet contract requirements.

* 4.3.2 Road test. The first production vehicle shall be examined and road tested for not less than 8 kilometers (km) (5 miles) without payload by the contractor to assure that the vehicle will operate in accordance with contract requirements.

4.3.3 Leakage test. The completely assembled tank body, including all piping, shall be tested for leaks at its design pressure in accordance with DOT Safety Regulations Governing Transportation of Hazardous Materials, Section 178.341, Specification MC 306. In addition, all piping and hoses shall be tested for leaks at the maximum pressure they can encounter during use of the vehicle-mounted pump.

4.3.4 Pump test. Vehicle pumping equipment shall be tested to determine performance requirements specified in 3.5.4.

4.3.6 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in an as new condition for the duration of the contract.

4.3.6 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. Each vehicle shall be tested for leakage in accordance with 4.3.3. For each vehicle under contract, the contractor shall make



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available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.4.24 have been accomplished.

* 5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The vehicle covered by this specification is intended for general administrative use for cleaning sewers with water at high pressure.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
* (b) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.2 and 2.2).
(c) Identification of appropriate service for painting and marking (see 3.1.1.1).
* (d) Concealed markings, if required (see 3.1.1.2).
(e) Rustproofing, if required (see 3.1.1.3).
(f) Tropical rustproofing, if required (see 3.1.1.3).
(g) Trailer towing package, if required (see 3.1.1.7).
* (h) Tool box, if required (see 3.1.1.12).
* (i) Satisfactory operation on JP-4, JP-5 and JP-8 fuels, if required (see 3.4.1.1).
(j) Gasoline engine driven vehicle. if required (see 3.4.1.2).
(k) Silicone rubber radiator and heater hoses. if required (see 3.4.1.8).
(l) Power plant heaters and fuel warmers, if required (see 3.4.1.9).
* (m) A fuel fired engine preheater, if required for a diesel engine driven vehicle (see 3.4.1.10).
* (n) Vernier throttle control, if required (see 3.4.1.11).
* (o) Starting motor circuit breaker, if required (see 3.4.2.1).
(p) Alternator capacity, if not as specified (see 3.4.2.3).
(q) A spark arrester, if required (see 3.4.4.1).
(r) Manual transmission, if required in lieu of automatic (see 3.4.5).
* (s) Auxiliary rear springs, if required (see 3.4.8).
(t) Two speed axle, if required (see 3.4.9.1).
(u) Traction control, if required (see 3.4.9.2).
(v) Disc type wheels, if required (see 3.4.10).



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(w) Bias ply tires, if required (see 3.4.10.1).
* (x) Low profile tires, if required (see 3.4.10.1).
(y) Carrier for a spare wheel or rim and tire assembly if required (see 3.4.10.3).
(z) Spare wheel or rim, if required (see 3.4.10.4).
(aa) Mounted spare tire, if required (see 3.4.10.5).
(ab) Trailer brake control system, if required (see 3.4.11.2).
(ac) Brake control from a towing vehicle, if required (see 3.4.11.2.1).
(ad) Tilt cab, if required (see 3.4.12).
(ae) Individual adjustable driver's and passenger seats, if required (see 3.4.12).
* (af) Intermittent windshield wipers, if required (3.4.14).
(ag) Tools, if required (see 3.4.16.1).
* (ah) Remote control curbside rearview mirrors, if required (see 3.4.20).
* (ai) Heated rearview mirrors, if required (see 3.4.20).
(aj) An air horn, in addition; if required (see 3.4.21).
* (ak) AM/FM radio, if required (see 3.4.23).
(al) Cooling system protection down to -54°C (-65°F), if required (see 3.4.24.1).

* 6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contractor. The SAE Work Sheet Item 1 should include vehicle model number and engine model number. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry, air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

| | |
|---|---|
| Table 1 | - Tire factor. This factor must relate to the size of the tires furnished by the contractor in accordance with this specification. |
| Table 2 | - Altitude factor. 1.00 |
| Table 3 | - Rolling factor. 1.613 |
| * Table 4 | - Area factor. 0.173; for truck tractors use 0.225. |
| Table 5 | - Velocity factor. 250.00 |
| Table 6 | - Altitude factor. 1.00. |
| Table 7 | - Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher 454 kg (1,000 pounds)) and the engine rpm (to nearest 100 revolutions) which is required for 80 km/h (50 mph) geared speed. For GVW and engine speed beyond the range of this table, factors shall be xtrapolated. |
| Table 8 | - Grade Factor. 0.75. |
| Table 8A | - Correction Factor. Not required. |
| Table 9 | - Road Factor. 0.0. |



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* 6.4 Subject term (key word) listing.

Auxiliary engine
Chassis, truck, commercial
High pressure cleaning gun
Hydraulic drive system
Nontactical truck
Tank
Water pump.

6.5 Identification of changes. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications. corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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